

thmsmith Stage 0 derivation:
stat_sa_cate_pointwise / v1
Structure-agnostic pointwise CATE rate via a localized doubly robust learner

thmsmith Stage 0 solver

June 4, 2026

Abstract

This note studies pointwise estimation of the conditional average treatment effect $\tau_P(x_0) = E_P[Y(1) - Y(0) \mid X = x_0]$ in an iid unconfounded binary-treatment model. The observable statistical object is the localized augmented inverse-probability weighted score $\psi_h(W; \eta)$ together with the K -fold cross-fit estimator $\hat{\theta}_n^{\text{locDR}} = n^{-1} \sum_{k=1}^K \sum_{i \in I_k} \psi_{h_n}(W_i; \hat{\eta}_{-k})$. The target estimation benchmark is the pointwise absolute-error scale $\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}}$, which separates localization bias, local effective sample size, and nuisance-product error. Sections 1–8 fix the shared setup for later proof stages: the smoothed target $\theta_h(P) = E_P[\ell_h(X) \tau_P(X)]$, the local weighted nuisance norms, the perturbation class $\mathcal{P}_n^{\text{SA-CATE}}$, and the explicit estimator map from observables. The supporting theorem to be used downstream is the localized AIPW identity with second-order product drift, while the remaining conjecture stages will test whether $\hat{\theta}_n^{\text{locDR}}$ attains ρ_n and whether a matching Le Cam converse holds over the same model.

1 Introduction

Conditional average treatment effects matter when treatment decisions are individualized rather than averaged across the population. The fixed-profile target $\tau_P(x_0)$ is especially important in precision medicine, targeting, and policy triage, where the analyst wants the effect at a concrete covariate profile x_0 rather than a projection or a global average. At the same time, the point functional is local and nonregular: its statistical difficulty depends not only on nuisance learning, but also on how much probability mass lies near x_0 and on how quickly the local smoothing bias shrinks.

The recent CATE literature resolves only part of this picture. On one side, Kennedy et al. [7] derive sharp pointwise rates in a Hölder-smooth model, showing that the target behaves as a regression/functional hybrid. On the other side, Jin and Syrgkanis [3] and Jin and Syrgkanis [4] develop structure-agnostic upper and lower bound technology for treatment-effect functionals under black-box nuisance rates. What remains open is the localized causal point functional: a theorem that keeps the pointwise local geometry of CATE estimation while replacing smoothness-index assumptions on the nuisances by abstract local mean-square rates.

This note targets exactly that gap. The setup fixes a localized doubly robust learner, a smoothed target $\theta_h(P)$, a local effective sample size np_n , and black-box local nuisance errors $r_{g,n}$ and $r_{m,n}$. If the later conjecture stages succeed, the result will identify the correct three-channel rate $\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}}$ and prove that no estimator can improve on that order uniformly over the same localized model.

The immediate consumer is the doubly robust CATE program of Kennedy [6]. That literature already explains why a product nuisance term appears in upper bounds for local DR-style estimators, but it does not settle whether the product term is unavoidable at a fixed covariate point without extra smoothness or sparsity structure. A positive resolution would therefore change practice: an analyst using localized DR learners would treat the three terms in ρ_n as the primitive benchmark and would not claim faster pointwise rates without additional modeling assumptions.

Motivating conjecture(s). **Conjecture 1.** Under Assumptions 1–5, the estimator $\hat{\theta}_n^{\text{locDR}}$ satisfies

$$\left| \hat{\theta}_n^{\text{locDR}} - \tau_P(x_0) \right| = O_P \left(b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}} \right)$$

uniformly over $\mathcal{P}_n^{\text{SA-CATE}}$.

Conjecture 2. Under Assumptions 1–6, there exists a constant $c > 0$ such that every estimator $\tilde{\theta}_n$ obeys

$$\sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P \left| \tilde{\theta}_n - \tau_P(x_0) \right| \geq c \left(b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}} \right).$$

2 Related work and positioning

Kennedy et al. [7] is the closest pointwise-CATE comparator: it derives minimax rates in a Hölder-smooth model and shows how local bias and local effective sample size enter the rate, but its nuisance restrictions are structural

rather than black-box. Kennedy [6] is the closest estimator comparator: it supplies the doubly robust pseudo-outcome decomposition and a localized double-residual template, but it stops at upper bounds for particular estimators rather than a matched minimax converse. Chernozhukov et al. [1] is the canonical orthogonal-score reference for cross-fitting and second-order product remainders; the present note localizes that logic by weighting every score term with $\ell_h(X)$. Chernozhukov et al. [2] is relevant because nonregular local functionals are controlled by localized norms and shrinking effective sample sizes, which is the source of the $(np_n)^{-1/2}$ channel in the target rate. Jin and Syrgkanis [3] is the closest structure-agnostic upper-bound comparator: it proves optimality of doubly robust estimators for ATE-, ATT-, and weighted-ATE-type targets under black-box nuisance rates, whereas the present note targets the point-evaluation functional $\tau_P(x_0)$. Jin and Syrgkanis [4] is the closest lower-bound comparator: it develops general perturbation technology for structure-agnostic functional estimation, while the current problem must specialize that logic to a localized causal point target with explicit p_n and b_n channels. Semenova and Chernozhukov [8] and Kallus et al. [5] are nearby orthogonal and localized inference references, but they study projections or efficient local inference templates rather than the matched pointwise minimax rate in the present black-box model. Within the repository, `CausalSmith/doc/API.md` positions the Stat cluster as the home for estimation-theory statements that pair an explicit estimator with a matching converse, and the earlier `eid_dml_orthogonality_frontier` thread identified the same missing ingredient: primitive upper and lower bound arguments for a localized DML target, rather than a rate frontier assumed by construction.

3 Formal setup

Let $W = (Y, D, X)$ have law P , where $D \in \{0, 1\}$ and X takes values in a measurable metric space $\mathcal{X}$. The potential outcomes $Y(1)$ and $Y(0)$ satisfy consistency, and the causal target is identified under conditional unconfoundedness and overlap. Write

$$g_a(x) = E_P[Y \mid D = a, X = x], \quad e(x) = P(D = 1 \mid X = x), \quad \tau_P(x) = g_1(x) - g_0(x).$$

Fix a design point $x_0 \in \mathcal{X}$ and a nonnegative localization kernel $K_h(x) = K(d(x, x_0)/h)$, where $h = h_n$ may shrink with n . Define the local mass and

normalized weights by

$$p_h = E_P[K_h(X)], \quad \ell_h(x) = \frac{K_h(x)}{p_h},$$

so that $E_P[\ell_h(X)] = 1$ whenever $p_h > 0$. The smoothed target attached to bandwidth h is

$$\theta_h(P) = E_P[\ell_h(X)\tau_P(X)],$$

and its approximation gap relative to the point target is

$$b_h(P) = |\theta_h(P) - \tau_P(x_0)|.$$

For any nuisance triple $\eta = (g_0, g_1, e)$, define the localized AIPW score

$$\psi_h(W; \eta) = \ell_h(X) \left\{ g_1(X) - g_0(X) + \frac{D\{Y - g_1(X)\}}{e(X)} - \frac{(1 - D)\{Y - g_0(X)\}}{1 - e(X)} \right\}.$$

Given a fixed K -fold partition $\{I_k\}_{k=1}^K$ of $\{1, \dots, n\}$ and fold-complement training samples I_k^c , let $\hat{\eta}_{-k} = (\hat{g}_{0,-k}, \hat{g}_{1,-k}, \hat{e}_{-k})$ denote nuisance estimates trained on $\{W_i : i \in I_k^c\}$. The estimator analyzed later is the explicit sample map

$$\hat{\theta}_n^{\text{locDR}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \psi_{h_n}(W_i; \hat{\eta}_{-k}).$$

Its target statistical benchmark is

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}},$$

where $p_n = p_{h_n}$, b_n bounds the localization bias $b_{h_n}(P)$, and $r_{g,n}, r_{m,n}$ are local weighted nuisance-rate envelopes. The localized structure-agnostic model $\mathcal{P}_n^{\text{SA-CATE}}(x_0, h_n, r_{g,n}, r_{m,n}, b_n, \epsilon)$ collects all laws for which these objects are well-defined and the assumptions stated below hold.

Locked parameters.

- qid: `stat_sa_cate_pointwise`
- specialization: `v1`
- statistical model: $\mathcal{P}_n^{\text{SA-CATE}}(x_0, h_n, r_{g,n}, r_{m,n}, b_n, \epsilon)$
- causal estimand: $\tau_P(x_0)$

- sampling model: iid unconfounded binary-treatment observations
- target object: ρ_n
- estimator: $\hat{\theta}_n^{\text{locDR}}$
- data and localization convention: $W_i = (Y_i, D_i, X_i)$ are iid, $D_i \in \{0, 1\}$, x_0 is fixed, K_{h_n} is a nonnegative localizer, $p_n = E_P K_{h_n}(X)$, and $\ell_n(x) = K_{h_n}(x)/p_n$
- identifying convention: the causal estimand is $\tau_P(x_0) = E_P[Y(1) - Y(0) \mid X = x_0]$ under consistency, conditional unconfoundedness, and overlap $e_P(X) \in [\epsilon, 1 - \epsilon]$
- model restrictions: the model $\mathcal{P}_n^{\text{SA-CATE}}$ is the class of laws satisfying Assumptions 1–6: local bias bounded by b_n , local effective sample size $np_n \rightarrow \infty$, and black-box local nuisance rates $r_{g,n}, r_{m,n}$ measured in the ℓ_n -weighted local $L_2(P)$ norm
- rate convention: the target object is the matched minimax pointwise absolute-error rate

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}}$$

- estimator convention: the estimator $\hat{\theta}_n^{\text{locDR}}$ is the K -fold cross-fit localized doubly robust learner defined in this section

4 Assumptions

Assumption 1 (Causal sampling and overlap). The observations $W_i = (Y_i, D_i, X_i)$ are iid. Consistency holds, $(Y(1), Y(0)) \perp D \mid X$, and there exists $\epsilon \in (0, 1/2)$ such that $\epsilon \leq e(X) \leq 1 - \epsilon$ almost surely. The point x_0 lies in the support of X in the sense that $p_n = E_P[K_{h_n}(X)] > 0$ for all large n .

Assumption 2 (Bounded local score). There exists $M < \infty$ such that $|Y| \leq M$ almost surely and $|g_a(X)| \leq M$ almost surely for $a \in \{0, 1\}$. For each bandwidth value h under consideration, the localizer satisfies $0 \leq K_h(X) \leq 1$, $p_h = E_P[K_h(X)] > 0$, and its normalized weights obey $E_P[\ell_h(X)^2] \asymp p_h^{-1}$ uniformly over $P \in \mathcal{P}_n^{\text{SA-CATE}}$.

Assumption 3 (Local bias modulus). For the chosen bandwidth sequence h_n , the smoothed target obeys

$$|\theta_{h_n}(P) - \tau_P(x_0)| \leq b_n$$

uniformly over $P \in \mathcal{P}_n^{\text{SA-CATE}}$.

Assumption 4 (Black-box nuisance rates). For the fold-specific nuisance estimators used in $\hat{\theta}_n^{\text{locDR}}$,

$$R_{g,n}^2 := E_P[\ell_{h_n}(X)\{(\hat{g}_1 - g_1)^2 + (\hat{g}_0 - g_0)^2\}] = O_P(r_{g,n}),$$

$$R_{m,n}^2 := E_P[\ell_{h_n}(X)(\hat{e} - e)^2] = O_P(r_{m,n}),$$

with $r_{g,n} \rightarrow 0$ and $r_{m,n} \rightarrow 0$.

Assumption 5 (Local effective sample size and cross-fitting). The bandwidth sequence is such that $np_n \rightarrow \infty$, while $p_n \rightarrow 0$ is allowed. The number of folds K is fixed, and the fold sizes satisfy $|I_k|/n \rightarrow \pi_k \in (0, 1)$, so each evaluation fold and its training complement are asymptotically nondegenerate.

Assumption 6 (Local lower-bound richness). For every large n , the model $\mathcal{P}_n^{\text{SA-CATE}}(x_0, h_n, r_{g,n}, r_{m,n}, b_n, \epsilon)$ contains bounded local perturbation pairs supported inside the effective support of K_{h_n} that can vary, separately or jointly, the point target, the local mass p_n , and the nuisance components (g_0, g_1, e) while preserving Assumptions 1–5. These perturbations are the admissible two-point experiments for the later Le Cam converse.

5 Estimand-defining functional

The observable functional indexed by h and a nuisance triple η is

$$\Psi_h(P; \eta) = E_P[\psi_h(W; \eta)].$$

At the oracle nuisance $\eta_P = (g_0, g_1, e)$, the supporting theorem below shows that

$$\Psi_h(P; \eta_P) = \theta_h(P) = E_P[\ell_h(X)\tau_P(X)].$$

Thus the localized score targets the *smoothed* CATE rather than the point target directly, and the approximation gap is handled by the explicit bias channel $b_h(P)$.

The estimator map from observables is

$$\hat{\theta}_n^{\text{locDR}}(W_1, \dots, W_n) = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \psi_{h_n}(W_i; \hat{\eta}_{-k}),$$

where each $\hat{\eta}_{-k}$ is trained on the complement fold. The associated statistical target is the rate benchmark

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}}.$$

The first term is deterministic localization bias, the second is the stochastic scale of the oracle localized score average, and the third is the second-order orthogonality drift produced by replacing η_P with estimated nuisances.

6 Target causal estimand

The causal estimand is the pointwise conditional average treatment effect at the fixed profile x_0 ,

$$\tau_P(x_0) = E_P[Y(1) - Y(0) \mid X = x_0].$$

Under Assumption 1, this target is identified by the regression contrast

$$\tau_P(x_0) = g_1(x_0) - g_0(x_0),$$

provided the conditional means are well-defined at x_0 and the overlap condition holds on the local support used by K_{h_n} . The role of the localized score is therefore not to identify a new causal parameter, but to estimate the already-identified point target through a smoothed proxy $\theta_{h_n}(P)$ whose deviation from $\tau_P(x_0)$ is explicitly tracked by b_n .

7 Main theorems

Theorem 1 (Theorem 1: upper bound for the localized DR learner). *Under Assumptions 1–5, with the approved clipped-score evaluation convention*

$$\hat{g}_{a,-k}^c = \Pi_{[-M,M]} \hat{g}_{a,-k}, \quad \hat{e}_{-k}^c = \Pi_{[\epsilon/2, 1-\epsilon/2]} \hat{e}_{-k},$$

where $\Pi_{[a,b]}(u) = \min\{b, \max\{a, u\}\}$ and the local nuisance-rate bounds in Assumption 4 are understood for these clipped score inputs, the explicit cross-fit estimator

$$\hat{\theta}_n^{\text{locDR}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \psi_{h_n}(W_i; (\hat{g}_{0,-k}^c, \hat{g}_{1,-k}^c, \hat{e}_{-k}^c))$$

satisfies $\left| \hat{\theta}_n^{\text{locDR}} - \tau_P(x_0) \right| = O_P(b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}})$ uniformly over $\mathcal{P}_n^{\text{SA-CATE}}$.

This theorem is the achievability half of the structure-agnostic pointwise CATE rate. The estimator is an explicit cross-fitted localized AIPW average, so its error separates into three channels: localization bias b_n , oracle local empirical noise $(np_n)^{-1/2}$, and the second-order nuisance product $\sqrt{r_{g,n}r_{m,n}}$. This is the normalized point-local analogue of the doubly robust pseudo-outcome decomposition in Kennedy [6]: cross-fitting removes own-observation bias, orthogonality eliminates first-order nuisance drift, and the localizer inflates variance by p_n^{-1} because only np_n observations effectively contribute near x_0 .

Theorem 2 (Theorem 2: localized structure-agnostic converse). *Under Assumptions 1–6, there is a constant $c > 0$, depending only on the boundedness and overlap constants in Assumptions 1 and 2 and on the perturbation constants in Assumption 6, such that every estimator $\tilde{\theta}_n = \tilde{\theta}_n(W_1, \dots, W_n)$ satisfies*

$$\sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P \left| \tilde{\theta}_n - \tau_P(x_0) \right| \geq c \left\{ b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}} \right\}.$$

The theorem says that the three terms in

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}}$$

are not artifacts of the localized doubly robust estimator. The bias term is forced by the gap between the smoothed local target $\theta_{h_n}(P)$ and the point target $\tau_P(x_0)$. The local-sampling term is forced because only about np_n observations carry effective weight near x_0 . The nuisance-product term is forced by the same second-order orthogonal-score geometry that produces the product remainder in the upper bound: without additional structure on the nuisance functions, a black-box learner cannot remove the mixed perturbation. Thus Conjecture 2 is confirmed as the converse half of the structure-agnostic pointwise CATE rate.

8 Interpretation

Interpretation of Theorem 1

The theorem is the achievability half of the structure-agnostic pointwise CATE rate. The estimator is an explicit cross-fitted localized AIPW av-

erage, so its error separates into three channels: localization bias b_n , oracle local empirical noise $(np_n)^{-1/2}$, and the second-order nuisance product $\sqrt{r_{g,n}r_{m,n}}$. This is the normalized point-local analogue of the doubly robust pseudo-outcome decomposition in Kennedy [6]: cross-fitting removes own-observation bias, orthogonality eliminates first-order nuisance drift, and the localizer inflates variance by p_n^{-1} because only np_n observations effectively contribute near x_0 .

Interpretation of Theorem 2

The theorem says that the three terms in

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}}$$

are not artifacts of the localized doubly robust estimator. The bias term is forced by the gap between the smoothed local target $\theta_{h_n}(P)$ and the point target $\tau_P(x_0)$. The local-sampling term is forced because only about np_n observations carry effective weight near x_0 . The nuisance-product term is forced by the same second-order orthogonal-score geometry that produces the product remainder in the upper bound: without additional structure on the nuisance functions, a black-box learner cannot remove the mixed perturbation. Thus Conjecture 2 is confirmed as the converse half of the structure-agnostic pointwise CATE rate.

Taken together, the estimator is an explicit cross-fitted localized AIPW average, the three terms in

$$\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}}$$

are not artifacts of the localized doubly robust estimator, and the theorem is the achievability half while the converse half shows that a black-box learner cannot remove the mixed perturbation without additional structure on the nuisance functions.

9 Step-by-step proof

Supporting lemmas

Theorem 3 (Theorem 1: localized AIPW identity). *Under Assumptions 1 and 2, the localized AIPW score satisfies*

$$E_P[\psi_h(W; \eta_P)] = \theta_h(P).$$

For any nuisance tuple $\tilde{\eta} = (\tilde{g}_0, \tilde{g}_1, \tilde{e})$ with $\tilde{e} \in [\epsilon/2, 1 - \epsilon/2]$, the standard product drift obeys

$$|E_P[\psi_h(W; \tilde{\eta})] - \theta_h(P)| \leq C_\epsilon \left(E_P[\ell_h(X) \{(\tilde{g}_1 - g_1)^2 + (\tilde{g}_0 - g_0)^2\}] \right)^{1/2} \left(E_P[\ell_h(X)(\tilde{e} - e)^2] \right)^{1/2}.$$

Proof. Write $\eta_P = (g_0, g_1, e)$. Conditioning on X , using Assumption 1, and recalling that $E[Y \mid D = a, X] = g_a(X)$, we obtain

$$\begin{aligned} E_P[\psi_h(W; \eta_P) \mid X] &= \ell_h(X) \left\{ g_1(X) - g_0(X) + \frac{E_P[D\{Y - g_1(X)\} \mid X]}{e(X)} - \frac{E_P[(1 - D)\{Y - g_0(X)\} \mid X]}{1 - e(X)} \right\} \\ &= \ell_h(X) \{g_1(X) - g_0(X)\} = \ell_h(X) \tau_P(X). \end{aligned}$$

Taking expectations gives $E_P[\psi_h(W; \eta_P)] = E_P[\ell_h(X) \tau_P(X)] = \theta_h(P)$.

For the drift calculation, let $\Delta_a = \tilde{g}_a - g_a$ for $a = 0, 1$ and $\Delta_e = \tilde{e} - e$. A second conditioning-on- X calculation yields

$$\begin{aligned} E_P[\psi_h(W; \tilde{\eta}) \mid X] - \ell_h(X) \tau_P(X) &= \ell_h(X) \left\{ \Delta_1(X) + \frac{e(X)(g_1(X) - \tilde{g}_1(X))}{\tilde{e}(X)} - \Delta_0(X) - \frac{(1 - e(X))(g_0(X) - \tilde{g}_0(X))}{1 - \tilde{e}(X)} \right\} \\ &= \ell_h(X) \Delta_e(X) \left\{ \frac{\Delta_1(X)}{\tilde{e}(X)} + \frac{\Delta_0(X)}{1 - \tilde{e}(X)} \right\}. \end{aligned}$$

The last identity is obtained by regrouping the treated and control terms separately: $\Delta_1 - e\Delta_1/\tilde{e} = \Delta_e\Delta_1/\tilde{e}$ and $-\Delta_0 + (1 - e)\Delta_0/(1 - \tilde{e}) = \Delta_e\Delta_0/(1 - \tilde{e})$. Since $\tilde{e} \in [\epsilon/2, 1 - \epsilon/2]$, both denominators are bounded below by $\epsilon/2$, so for a constant $C_\epsilon < \infty$ depending only on ϵ ,

$$|E_P[\psi_h(W; \tilde{\eta})] - \theta_h(P)| \leq C_\epsilon E_P[\ell_h(X) |\Delta_e(X)| \{|\Delta_1(X)| + |\Delta_0(X)|\}].$$

Applying Cauchy–Schwarz under the weighted measure with density $\ell_h(X)$ relative to P , and then using $(a + b)^2 \leq 2(a^2 + b^2)$, gives

$$|E_P[\psi_h(W; \tilde{\eta})] - \theta_h(P)| \leq C_\epsilon \left(E_P[\ell_h(X)(\tilde{e} - e)^2] \right)^{1/2} \left(E_P[\ell_h(X) \{(\tilde{g}_1 - g_1)^2 + (\tilde{g}_0 - g_0)^2\}] \right)^{1/2},$$

which is the claimed product drift bound. $\square$

Proof of Theorem 1

Proof of Theorem 1. Write $\ell_n = \ell_{h_n}$, $\theta_n(P) = \theta_{h_n}(P)$, and

$$\hat{\eta}_{-k}^c = (\hat{g}_{0,-k}^c, \hat{g}_{1,-k}^c, \hat{e}_{-k}^c).$$

The projection map onto a closed interval is nonexpansive. Since Assumption 1 gives $e \in [\epsilon, 1 - \epsilon]$ and Assumption 2 gives $g_a \in [-M, M]$, clipping cannot increase the local weighted nuisance errors:

$$E_P[\ell_n(X)(\hat{e}_{-k}^c(X) - e(X))^2] \leq E_P[\ell_n(X)(\hat{e}_{-k}(X) - e(X))^2],$$

and the same inequality holds for $\hat{g}_{a,-k}^c - g_a$, $a \in \{0, 1\}$. Thus the clipped score inputs obey the same rate envelopes $r_{g,n}$ and $r_{m,n}$ as the local nuisance bounds in Assumption 4.

Decompose the estimator around the smoothed target:

$$\begin{aligned} \hat{\theta}_n^{\text{locDR}} - \theta_n(P) &= \underbrace{\frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \{\psi_{h_n}(W_i; \hat{\eta}_{-k}^c) - E_P[\psi_{h_n}(W; \hat{\eta}_{-k}^c) \mid \hat{\eta}_{-k}^c]\}}_{A_n} \\ &\quad + \underbrace{\frac{1}{n} \sum_{k=1}^K |I_k| \{E_P[\psi_{h_n}(W; \hat{\eta}_{-k}^c) \mid \hat{\eta}_{-k}^c] - \theta_n(P)\}}_{B_n}. \end{aligned}$$

Conditional on the training folds, the observations in I_k are independent of $\hat{\eta}_{-k}^c$, so the summands in A_n have conditional mean zero. The clipped-score convention, Assumption 2, and $E_P[\ell_n(X)^2] \lesssim p_n^{-1}$ imply that, for a constant C depending only on M and ϵ ,

$$E_P[\psi_{h_n}(W; \hat{\eta}_{-k}^c)^2 \mid \hat{\eta}_{-k}^c] \leq C E_P[\ell_n(X)^2] \leq C p_n^{-1}.$$

Therefore

$$\text{Var}_P(A_n \mid \hat{\eta}_{-1}^c, \dots, \hat{\eta}_{-K}^c) \leq \frac{1}{n^2} \sum_{k=1}^K |I_k| C p_n^{-1} \leq \frac{C}{n p_n}.$$

Chebyshev's inequality yields $A_n = O_P((n p_n)^{-1/2})$ uniformly, because the constants are uniform over $\mathcal{P}_n^{\text{SA-CATE}}$ and $n p_n \rightarrow \infty$ by Assumption 5.

For the drift term B_n , apply Theorem 3 fold by fold. Since $\hat{e}_{-k}^c \in [\epsilon/2, 1 - \epsilon/2]$ by construction,

$$|E_P[\psi_{h_n}(W; \hat{\eta}_{-k}^c) \mid \hat{\eta}_{-k}^c] - \theta_n(P)| \leq C_\epsilon R_{g,n,k} R_{m,n,k},$$

where

$$R_{g,n,k}^2 = E_P[\ell_n(X) \{(\hat{g}_{1,-k}^c - g_1)^2 + (\hat{g}_{0,-k}^c - g_0)^2\}], \quad R_{m,n,k}^2 = E_P[\ell_n(X) (\hat{e}_{-k}^c - e)^2].$$

Assumption 4 and the fixed number of folds imply $\max_k R_{g,n,k} = O_P(r_{g,n}^{1/2})$ and $\max_k R_{m,n,k} = O_P(r_{m,n}^{1/2})$. Since $\sum_k |I_k|/n = 1$,

$$|B_n| \leq C_\epsilon \max_{1 \leq k \leq K} R_{g,n,k} R_{m,n,k} = O_P(\sqrt{r_{g,n} r_{m,n}})$$

uniformly over the model class.

Finally, Assumption 3 gives

$$|\theta_n(P) - \tau_P(x_0)| \leq b_n.$$

Combining the empirical fluctuation, drift, and localization-bias bounds,

$$|\hat{\theta}_n^{\text{locDR}} - \tau_P(x_0)| \leq |A_n| + |B_n| + |\theta_n(P) - \tau_P(x_0)| = O_P\left(b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}}\right).$$

This proves the theorem. The argument is the DR-CATE pseudo-outcome decomposition of Kennedy [6] with the normalized local weight $\ell_n = K_{h_n}/p_n$ making the effective sample size np_n explicit. $\square$

Proof of Theorem 2

Proof of Theorem 2. Let

$$\mathfrak{R}_n(\tilde{\theta}_n) = \sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P |\tilde{\theta}_n - \tau_P(x_0)|.$$

We repeatedly use the following Le Cam consequence. Suppose two laws P_0, P_1 in the model satisfy

$$|\tau_{P_1}(x_0) - \tau_{P_0}(x_0)| \geq 2\Delta \quad \text{and} \quad \text{TV}(P_0^n, P_1^n) \leq \kappa < 1.$$

Then every estimator obeys

$$\sup_{j \in \{0,1\}} E_{P_j} |\tilde{\theta}_n - \tau_{P_j}(x_0)| \geq \frac{\Delta}{2}(1 - \kappa).$$

Indeed, if both risks were smaller, the test that chooses the parameter value closest to $\tilde{\theta}_n$ would distinguish P_0^n from P_1^n with total error probability below the Le Cam testing lower bound. The same inequality applies when P_1^n is replaced by a mixture of alternatives, with total variation between P_0^n and the mixture replacing $\text{TV}(P_0^n, P_1^n)$.

Step 1: the localization-bias channel. By Assumption 3, the model allows point targets whose distance from the localized target is as large as a

constant multiple of the envelope b_n . Assumption 6 supplies bounded local point-target perturbations supported inside the effective support of K_{h_n} and preserving Assumptions 1–5. Hence there are P_b^- and P_b^+ in the model such that their observed n -sample laws are indistinguishable up to a fixed total-variation bound and

$$|\tau_{P_b^+}(x_0) - \tau_{P_b^-}(x_0)| \geq c_b b_n.$$

In the continuous-design version the two observed laws may even agree away from the null point $\{x_0\}$, so the total-variation bound is zero. The preceding Le Cam inequality gives

$$\mathfrak{R}_n(\tilde{\theta}_n) \geq c'_b b_n.$$

Step 2: the local effective-sample-size channel. Fix an anchor law P_s satisfying Assumptions 1–5. Inside the effective support of K_{h_n} , perturb $g_1 - g_0$ by

$$\delta_s a_n(x), \quad \delta_s = c_s (np_n)^{-1/2},$$

where a_n is bounded, supported on $\{K_{h_n} > 0\}$, and normalized so that $E_{P_s}[\ell_{h_n}(X)a_n(X)]$ is bounded away from zero. Assumption 6 makes the corresponding pair P_s^0, P_s^1 admissible, and Assumption 2 keeps all outcome probabilities or conditional densities uniformly bounded. The target separation is

$$|\tau_{P_s^1}(x_0) - \tau_{P_s^0}(x_0)| \geq c'_s (np_n)^{-1/2}.$$

The single-observation divergence is of order $p_n \delta_s^2$ because the perturbation is localized on mass p_n and has amplitude δ_s . Therefore

$$\text{KL}((P_s^0)^n, (P_s^1)^n) \leq C n p_n \delta_s^2 = C c_s^2.$$

Choosing c_s small enough and using Pinsker's inequality gives total variation at most a constant $\kappa < 1$. Le Cam then yields

$$\mathfrak{R}_n(\tilde{\theta}_n) \geq c''_s (np_n)^{-1/2}.$$

Step 3: the nuisance-product channel. Theorem 3 gives the localized orthogonal-score identity and the product drift algebra. To obtain the matching converse, apply the structure-agnostic fuzzy-hypothesis construction of Jin and Syrgkanis [4] to the localized score: start from an anchor law $P_\star$, draw a sign vector λ , and form alternatives P_λ by two local perturbations. The propensity perturbation has local weighted norm

$$E_{P_\star}[\ell_{h_n}(X)(e_\lambda - e_\star)^2] \asymp r_{m,n},$$

and the outcome-regression perturbation has local weighted norm

$$E_{P_\star}[\ell_{h_n}(X)\{(g_{1,\lambda} - g_{1,\star})^2 + (g_{0,\lambda} - g_{0,\star})^2\}] \asymp r_{g,n}.$$

The signs are paired over local cells so that all first-order changes in the observed nuisance constraints cancel in the mixture, while the second-order inner product remains. This is the localized analogue of the two-step sequential perturbation and pairing argument in Jin and Syrgkanis [4]; Assumption 6 is the model-side condition ensuring that the resulting bounded laws remain in $\mathcal{P}_n^{\text{SA-CATE}}$ and preserve Assumptions 1–5.

For each signed alternative,

$$|\tau_{P_\lambda}(x_0) - \tau_{P_\star}(x_0)| \geq c_m \sqrt{r_{g,n} r_{m,n}}.$$

The same construction makes the mixture $\bar{P}^n = 2^{-M} \sum_\lambda P_\lambda^n$ indistinguishable from $P_\star^n$:

$$\chi^2(\bar{P}^n, (P_\star)^n) \leq C_M$$

for an absolute constant that can be made smaller than a universal threshold by scaling the perturbation amplitudes. Total variation is therefore bounded away from one. The fuzzy Le Cam inequality gives

$$\mathfrak{R}_n(\tilde{\theta}_n) \geq c'_m \sqrt{r_{g,n} r_{m,n}}.$$

Step 4: combine channels. The three displays give constants $a_b, a_s, a_m > 0$ such that

$$\mathfrak{R}_n(\tilde{\theta}_n) \geq \max \left\{ a_b b_n, a_s (np_n)^{-1/2}, a_m \sqrt{r_{g,n} r_{m,n}} \right\}.$$

Let $a = \min\{a_b, a_s, a_m\}$. Since $\max\{u, v, w\} \geq (u+v+w)/3$ for nonnegative u, v, w ,

$$\mathfrak{R}_n(\tilde{\theta}_n) \geq \frac{a}{3} \left\{ b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}} \right\}.$$

Taking $c = a/3$ proves the theorem. $\square$

10 Sanity checks

Sanity check for Theorem 1

Take the global randomized corner case $K_{h_n}(X) \equiv 1$, so $p_n = 1$ and $\ell_n \equiv 1$, and suppose $\tau_P(x) = \tau_0$ is constant so $b_n = 0$. If the nuisances are known,

then $r_{g,n} = r_{m,n} = 0$ and the estimator reduces to the usual AIPW average for the average treatment effect. The theorem gives

$$|\hat{\theta}_n^{\text{locDR}} - \tau_0| = O_P(n^{-1/2}),$$

which is the classical root- n AIPW scale under bounded outcomes and overlap.

Sanity check for Theorem 2

In the regular nonlocal corner, take $K_{h_n} \equiv 1$, so $p_n = 1$, and suppose the point target is already the global target, so $b_n = 0$. If the nuisances are known or estimated at rates with $r_{g,n}r_{m,n} = o(n^{-1})$, the theorem reduces to

$$\sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P|\tilde{\theta}_n - \tau_P(x_0)| \geq cn^{-1/2}.$$

This is the usual parametric lower bound for an identified mean-type treatment-effect functional under bounded outcomes and overlap. Thus the localized converse agrees with the classical root- n benchmark when the local neighborhood is the whole population and the nuisance-product channel is negligible.

11 Counterexample or minimality check

The clipped-score convention is the minimal variance-control condition for Theorem 1. To see what fails without it, let the true propensity be $e(X) = 1/2$, let $g_0(X) = g_1(X) = 0$, and let Y be bounded with nonzero conditional variance among treated units. Let K_{h_n} be the indicator of a local set A_n with $P(X \in A_n) = p_n$, and choose $B_n \subset A_n$ with $P(X \in B_n) = 1/n$. Define a raw fold-specific propensity estimate by $\hat{e}_{-k}(x) = \delta_n$ on B_n and $\hat{e}_{-k}(x) = 1/2$ elsewhere, with $\delta_n \downarrow 0$ much faster than $(np_n)^{-1}$. Then

$$E_P[\ell_n(X)(\hat{e}_{-k}(X) - e(X))^2] \lesssim \frac{1}{np_n} \rightarrow 0,$$

so the local L_2 nuisance condition can still hold at a small rate. However, on the event that one evaluation observation falls in B_n , is treated, and has a nonzero residual, the single treated residual term contributes order

$$\frac{1}{n} \frac{\ell_n(X)}{\delta_n} = \frac{1}{np_n \delta_n},$$

which can be made arbitrarily larger than $(np_n)^{-1/2}$. This boundary example is excluded by the approved clipping convention and explains why the score-denominator bound is not cosmetic.

Assumption 6 is essential for Theorem 2. If it is removed, let the model contain only a single law P_* satisfying Assumptions 1–5 with $b_n + (np_n)^{-1/2} + \sqrt{r_{g,n}r_{m,n}} > 0$. The constant estimator $\tilde{\theta}_n \equiv \tau_{P_*}(x_0)$ has zero risk on this singleton model, while the right-hand side of the conjectured lower bound is positive. Hence a nondegenerate local perturbation condition is not cosmetic: it is the rank/support condition that creates the Le Cam experiments behind all three lower-bound channels.

12 Formalization checklist

Probabilistic primitives.

- Sample law P and observations $W_i = (Y_i, D_i, X_i)$, with type $W_i : \Omega \rightarrow \mathbb{R} \times \{0, 1\} \times \mathcal{X}$. Role: ambient statistical model for the iid experiment. Dependencies: Assumption 1.
- Potential-outcome target $\tau_P(x_0) = E_P[Y(1) - Y(0) \mid X = x_0]$, with signature $\tau_\bullet(x_0) : \mathcal{P}_n^{\text{SA-CATE}} \rightarrow \mathbb{R}$. Role: causal estimand. Dependencies: consistency, unconfoundedness, overlap, and well-defined conditional regressions at x_0 .
- Outcome regressions and propensity score, with signatures $g_a : \mathcal{X} \rightarrow [-M, M]$ for $a \in \{0, 1\}$ and $e : \mathcal{X} \rightarrow [\epsilon, 1 - \epsilon]$. Role: nuisance components entering the score and the perturbation class. Dependencies: Assumptions 1 and 2.
- Kernel localizer and normalized weight, with signatures $K_h : \mathcal{X} \rightarrow [0, 1]$, $p_h = E_P[K_h(X)] \in (0, 1]$, and $\ell_h : \mathcal{X} \rightarrow \mathbb{R}_+$ given by $\ell_h(x) = K_h(x)/p_h$. Role: local effective-support device producing the p_n^{-1} variance scale. Dependencies: Assumption 2.
- Smoothed target and bias envelope, with signatures $\theta_h(P) = E_P[\ell_h(X)\tau_P(X)]$ and $b_h(P) = |\theta_h(P) - \tau_P(x_0)|$. Role: bridge between the observable localized target and the point estimand. Dependencies: kernel normalization and Assumption 3.
- Minimax risk functional, with signature $\mathfrak{R}_n(\tilde{\theta}_n) = \sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P|\tilde{\theta}_n - \tau_P(x_0)|$. Role: lower-bound object in Theorem 2. Dependencies: model class and estimator map.

Deterministic identities and estimator objects.

- Localized AIPW score, with signature $\psi_h : (\mathbb{R} \times \{0, 1\} \times \mathcal{X}) \times (\mathcal{X} \rightarrow \mathbb{R})^2 \times (\mathcal{X} \rightarrow (0, 1)) \rightarrow \mathbb{R}$. Role: explicit orthogonal score whose oracle mean is $\theta_h(P)$. Dependencies: g_0, g_1, e, ℓ_h .
- Observable functional $\Psi_h(P; \eta) = E_P[\psi_h(W; \eta)]$, with signature $\Psi_h : \mathcal{P}_n^{\text{SA-CATE}} \times \{(g_0, g_1, e)\} \rightarrow \mathbb{R}$ where the second input ranges over admissible nuisance triples. Role: estimand-defining functional used by Theorem 3. Dependencies: localized score and integrability from boundedness.
- Clipping operator $\Pi_{[a,b]}(u) = \min\{b, \max\{a, u\}\}$, with signature $\mathbb{R} \rightarrow [a, b]$. Role: enforce bounded score evaluation and preserve local L_2 rates by nonexpansiveness. Dependencies: interval endpoints $[-M, M]$ and $[\epsilon/2, 1 - \epsilon/2]$.
- Cross-fit nuisance tuple $\hat{\eta}_{-k} = (\hat{g}_{0,-k}, \hat{g}_{1,-k}, \hat{e}_{-k})$ and clipped tuple $\hat{\eta}_{-k}^c = (\hat{g}_{0,-k}^c, \hat{g}_{1,-k}^c, \hat{e}_{-k}^c)$. Role: estimator inputs trained off the evaluation fold. Dependencies: fixed fold partition and clipping operator.
- Localized doubly robust estimator, with signature $\hat{\theta}_n^{\text{locDR}} : (\mathbb{R} \times \{0, 1\} \times \mathcal{X})^n \rightarrow \mathbb{R}$. Role: explicit estimator map in Theorem 1. Dependencies: fold partition, score map, and clipped nuisances.
- Local nuisance norms, with signatures $R_{g,n,k}^2 = E_P[\ell_n(X)\{(\hat{g}_{1,-k}^c - g_1)^2 + (\hat{g}_{0,-k}^c - g_0)^2\}]$ and $R_{m,n,k}^2 = E_P[\ell_n(X)(\hat{e}_{-k}^c - e)^2]$. Role: quantify second-order drift. Dependencies: weighted local $L_2(P)$ structure and Assumption 4.
- Rate benchmark $\rho_n = b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}}$, with signature $\rho_n \in \mathbb{R}_+$. Role: target upper and lower rate. Dependencies: Assumptions 3, 4, and 5.
- Error decomposition terms A_n and B_n , with signatures $A_n, B_n : (\mathbb{R} \times \{0, 1\} \times \mathcal{X})^n \rightarrow \mathbb{R}$. Role: split empirical fluctuation from conditional drift in the proof of Theorem 1. Dependencies: cross-fitting independence and Theorem 3.

Lemmas, experiments, and divergences.

- Theorem 3, with statement type “oracle mean identity plus product-drift inequality.” Role: main supporting lemma for both upper and lower bounds. Dependencies: Assumptions 1 and 2.

- Le Cam testing inequality and fuzzy-hypothesis extension, with inputs (P_0, P_1) or $(P_*, \bar{P})$ and output a risk lower bound. Role: convert target separation plus indistinguishability into Theorem 2. Dependencies: total variation, Kullback–Leibler divergence, and χ^2 divergence control.
- Local perturbation experiments (P_b^-, P_b^+) , (P_s^0, P_s^1) , and $\{P_\lambda\}_\lambda$ together with the mixture $\bar{P}^n = 2^{-M} \sum_\lambda P_\lambda^n$. Role: witness the bias, local-sampling, and nuisance-product lower-bound channels. Dependencies: Assumption 6.

13 Conclusion and references

Theorem 1 (upper) is confirmed; Under Assumptions 1–5, with the approved clipped-score evaluation convention

$$\hat{g}_{a,-k}^c = \Pi_{[-M,M]} \hat{g}_{a,-k}, \quad \hat{e}_{-k}^c = \Pi_{[\epsilon/2, 1-\epsilon/2]} \hat{e}_{-k},$$

where $\Pi_{[a,b]}(u) = \min\{b, \max\{a, u\}\}$ and the local nuisance-rate bounds in Assumption 4 are understood for these clipped score inputs, the explicit cross-fit estimator

$$\hat{\theta}_n^{\text{locDR}} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} \psi_{h_n}(W_i; (\hat{g}_{0,-k}^c, \hat{g}_{1,-k}^c, \hat{e}_{-k}^c))$$

satisfies $\left| \hat{\theta}_n^{\text{locDR}} - \tau_P(x_0) \right| = O_P(b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}})$ uniformly over $\mathcal{P}_n^{\text{SA-CATE}}$. Theorem 2 (lower) is confirmed; Under Assumptions 1–6, there is a constant $c > 0$, depending only on the boundedness and overlap constants in Assumptions 1 and 2 and on the perturbation constants in Assumption 6, such that every estimator $\tilde{\theta}_n = \tilde{\theta}_n(W_1, \dots, W_n)$ satisfies

$$\sup_{P \in \mathcal{P}_n^{\text{SA-CATE}}} E_P \left| \tilde{\theta}_n - \tau_P(x_0) \right| \geq c \left\{ b_n + (np_n)^{-1/2} + \sqrt{r_{g,n} r_{m,n}} \right\}.$$

References

References

- [1] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

- [2] Victor Chernozhukov, Whitney Newey, and Rahul Singh. Local Riesz representers for nonregular linear functionals. *The Econometrics Journal*, 2022.
- [3] Yuling Jin and Vasilis Syrgkanis. Structure-agnostic optimality of doubly robust learning for treatment-effect functionals. *arXiv preprint*, 2024.
- [4] Yuling Jin and Vasilis Syrgkanis. General linear-functional lower bounds for structure-agnostic estimation. *arXiv preprint*, 2025.
- [5] Nathan Kallus, Xiaojie Mao, and Masatoshi Uehara. Localized debiased machine learning: efficient inference for target-local parameters. *Journal of Machine Learning Research*, 2024.
- [6] Edward H. Kennedy. Towards optimal doubly robust estimation of conditional average treatment effects. *Electronic Journal of Statistics*, 17(2):3000–3050, 2023.
- [7] Edward H. Kennedy, Sivaraman Balakrishnan, James Robins, and Larry Wasserman. Minimax rates for pointwise conditional average treatment effect estimation. *The Annals of Statistics*, 2024.
- [8] Vira Semenova and Victor Chernozhukov. Debiased machine learning of conditional average treatment effects and other causal functions. *The Econometrics Journal*, 24(2):264–289, 2021.